

Lecture 3: Data Collection 2: Application Programming Interfaces (APIs)

IS 618: Social Media Data Analysis
Indira Sen, FSS 2026

1. (*neologism*, *Internet*, *vulgar*) The *phenomenon* of *online platforms* gradually *degrading* the quality of their *services*, often by promoting *advertisements*, *recurring payments*, and *sponsored content*, in order to increase *profits*. [synonyms, antonyms ▲] [quotations ▼]

Noun [\[edit \]](#)

enshittification (*uncountable*)

1. (*neologism*, *Internet*, *vulgar*) The *phenomenon* of *online platforms* gradually *degrading* the quality of their *services*, often by promoting *advertisements*, *recurring payments*, and *sponsored content*, in order to increase *profits*. [synonyms, antonyms ▲] [quotations ▼]

<https://en.wiktionary.org/wiki/enshittification#English>

Recap

- Pitfalls of studying society with social media
- What are the components of social media platforms?

The End of Theory: The Data Deluge Makes the **Scientific Method** Obsolete

By Chris Anderson 06.23.08

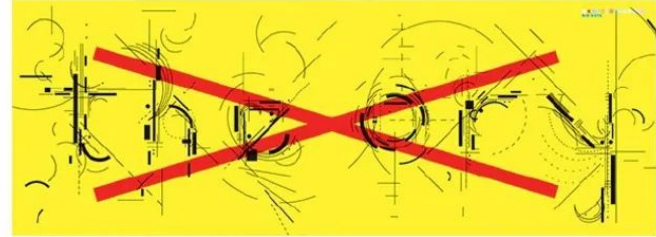


Illustration: Marian Bantjes

"All models are wrong, but some are useful."

So proclaimed statistician George Box 30 years ago, and he was right. But what choice did we have? Only models, from cosmological equations to theories of human behavior, seemed to be able to consistently, if imperfectly, explain the world around us. Until now. Today companies like Google, which have grown up in an era of massively abundant data, don't have to settle for wrong models. Indeed, they don't have to settle for models at all.

Sixty years ago, digital computers made information readable. Twenty years ago, the Internet made it reachable. Ten years ago, the first search engine crawlers made it a single database. Now Google and like-minded companies are sifting through the most measured age in history, treating this massive corpus as a laboratory of the human condition. They are the children of the Petabyte Age.

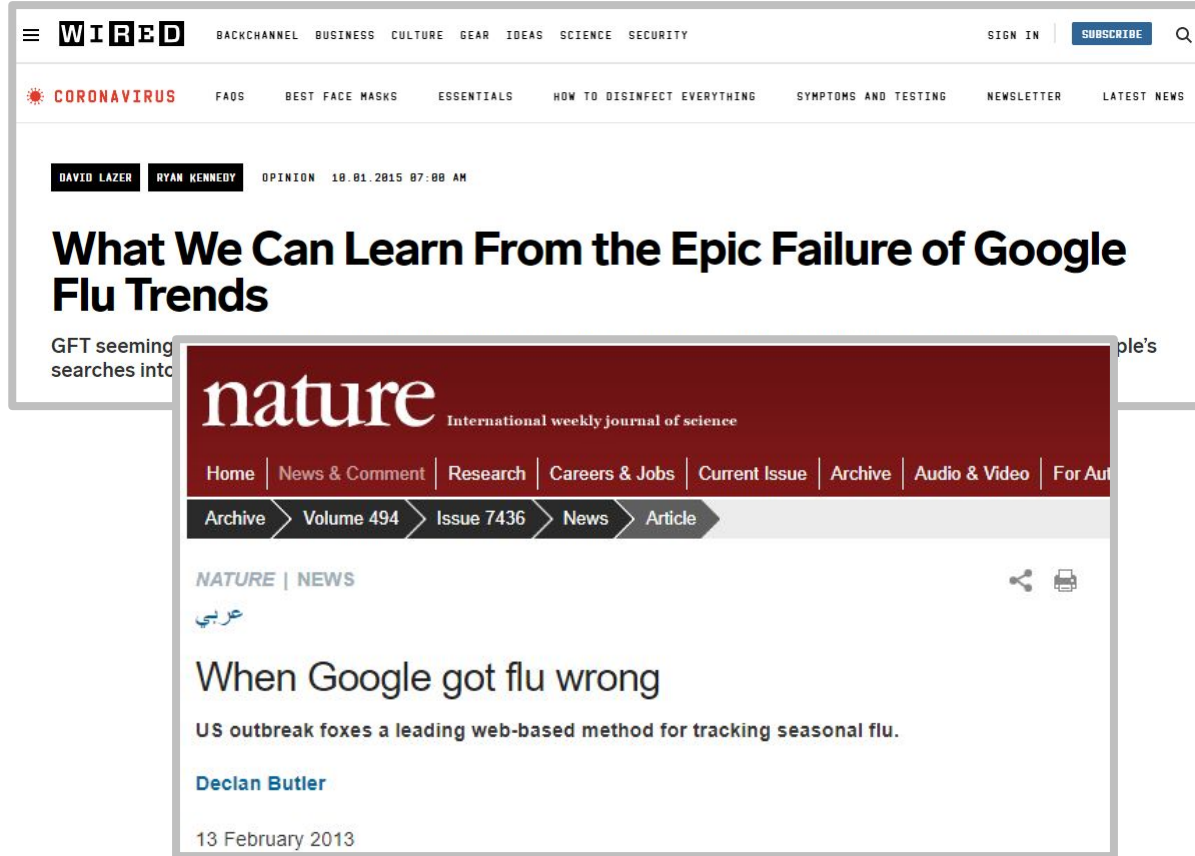
The Petabyte Age is different because more is different. Kilobytes were stored on floppy disks. Megabytes were stored on hard disks. Terabytes were stored in disk arrays. Petabytes are stored in the cloud. As we moved along that progression, we went from the folder analogy to the file cabinet analogy to the library analogy to — well, at petabytes we ran out of organizational analogies.

At the petabyte scale, information is not a matter of simple three- and four-dimensional taxonomy and order but of dimensionally agnostic statistics. It calls for an entirely different <https://www.wired.com/2008/06/pb-theory/> ^{e visualized in t for it later.}

For instance, Google conquered the advertising world with nothing more than applied mathematics. It didn't pretend to know anything about the culture and conventions of

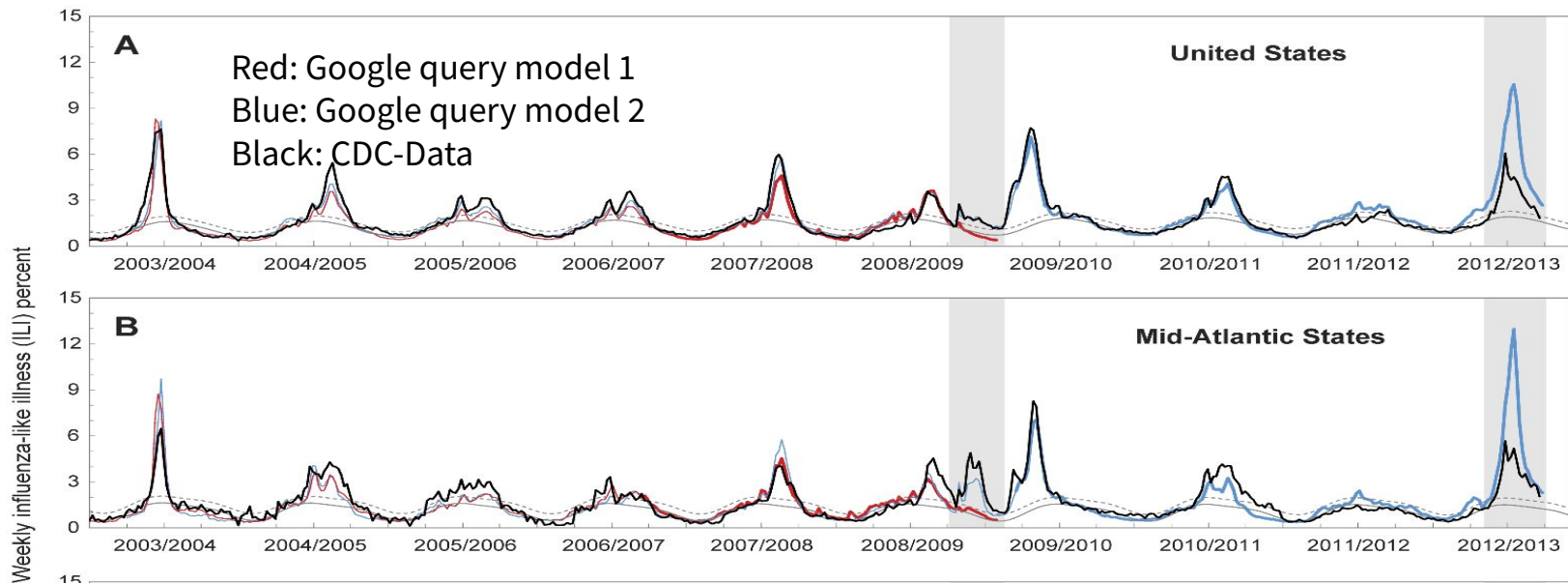
Studying Social Media: No Big Deal?

The fail



“In February 2013, Google Flu Trends (GFT) made headlines but not for a reason that Google executives or the creators of the flu tracking system would have hoped. Nature reported that GFT was predicting more than double the proportion of doctor visits for influenza-like illness (ILI) than the Centers for Disease Control and Prevention (CDC), which bases its estimates on surveillance reports from laboratories across the United States (1, 2).”

Overestimation



“part flue detector, part winter detector” □ errors are auto-correlated & direction and magnitude change with seasons

Where to be careful

“[...] there are enormous scientific possibilities in big data [but] the core challenge is that most big data that have received popular attention are not the output of instruments designed to produce valid and reliable data amenable for scientific analysis.”

“**Big data hubris** is the often implicit assumption that big data are a substitute for, rather than a supplement to, traditional data collection and analysis”

Lazer, David, et al. "[The parable of Google Flu: traps in big data analysis.](#)" *Science* 343.6176 (2014)

Social Media Data Analysis

The Good 🤗

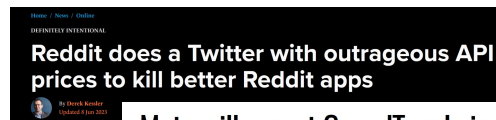
- Social Media provide a nice **complementary** data source to traditional social science data ([Lazer et al. 2009](#), [2020](#))
- The platforms themselves are an object of study
- Large scale data => an opportunity for data-driven approaches, like natural language processing (NLP)

The Bad 😞

- Disproportionate focus on particular platforms / sources ([Tufekci 2014](#))
- Many representative biases ([Olteanu et al., 2021](#))
- Ad-hoc measurement models (“what are we *really* measuring?”)
- Need for a measurement theory ([Jungherr 2018](#))

The Ugly 🤢

- The API Apocalypse ([Freelon 2018](#); [Assenmacher et al., 2023](#))



Meta will sunset CrowdTangle in August

Advertisers and publishers will need to seek alternative platforms as CrowdTangle will no longer be available after August 14.

SCIENCE / TWITTER - X / TECH

Twitter just closed the book on academic research



/ Twitter was once an indispensable resource for academic research. That's changed under Elon Musk.

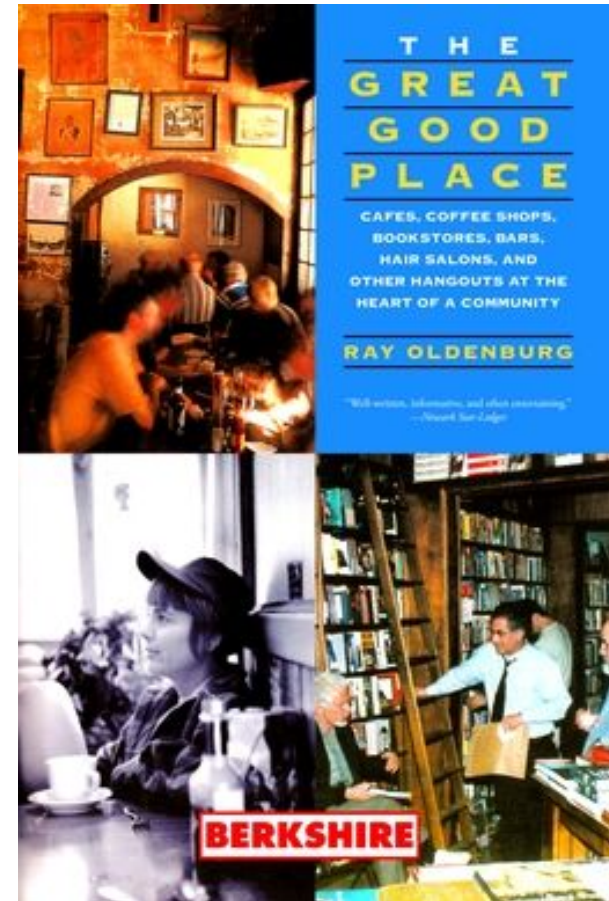


More on this next week

A Brief History of Social Media and the Elements of Modern Platforms

Why do we need social media?

- **Third Places:** “social surroundings that are separate from the two usual social environments of home (“first place”) and workplace (“second place”)”
 - Cafes, libraries, churches, vereins, ...
- Helps create a sense of community
- Physical third places have declined
- Social Media Platforms could be an alternative (but are they?) [[Bruckman 2022](#)]



[Oldenburg, 1989](#)

Categorizing Different Social Media Platforms

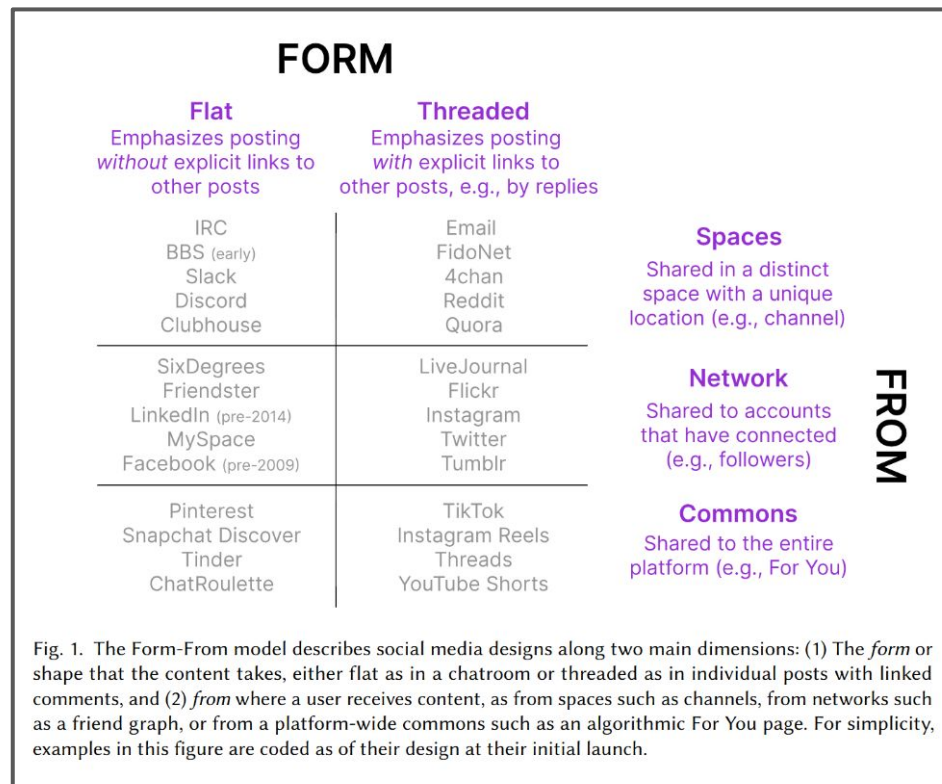
Some possible categorizations:

- **Ephemerality:** Snapchat (in early days) and stories vs. Facebook posts
- **Modality:** primarily text-based, e.g., Reddit, vs. Image-based e.g., Pinterest
- **Interaction type and timing:** e.g., BeReal
- **Moderation type:** Volunteer-run, e.g., Reddit vs. Commercial e.g., Facebook
- **Professional vs. personal use:** Facebook vs. LinkedIn
- **International or local:** Jappy, Pixnet
- **Value-based:** 'Free-speech' platforms like Truth Social

https://en.wikipedia.org/wiki/List_of_social_networking_services

Categorizing Different Social Media Platforms

- Form: the shape of the content:
 - **threaded** by emphasizing explicitly linking posts to each other
 - **flat** where the design emphasizes posts that are not explicitly linked to each other,
- From: content delivery
 - to the user from **spaces** (e.g., channels, groups, or rooms) where all members in the space receive it,
 - from (social) **networks** where each user independently curates their own network
 - a sample from the “**commons**” of all content on the platform?



The Elements of Modern Social Media Platforms

- The platform itself
- User base
 - Non-human users
 - User tiers
- Content
- Algorithms
 - Recommendation
 - Ranking
 - Moderation
- Advertisements

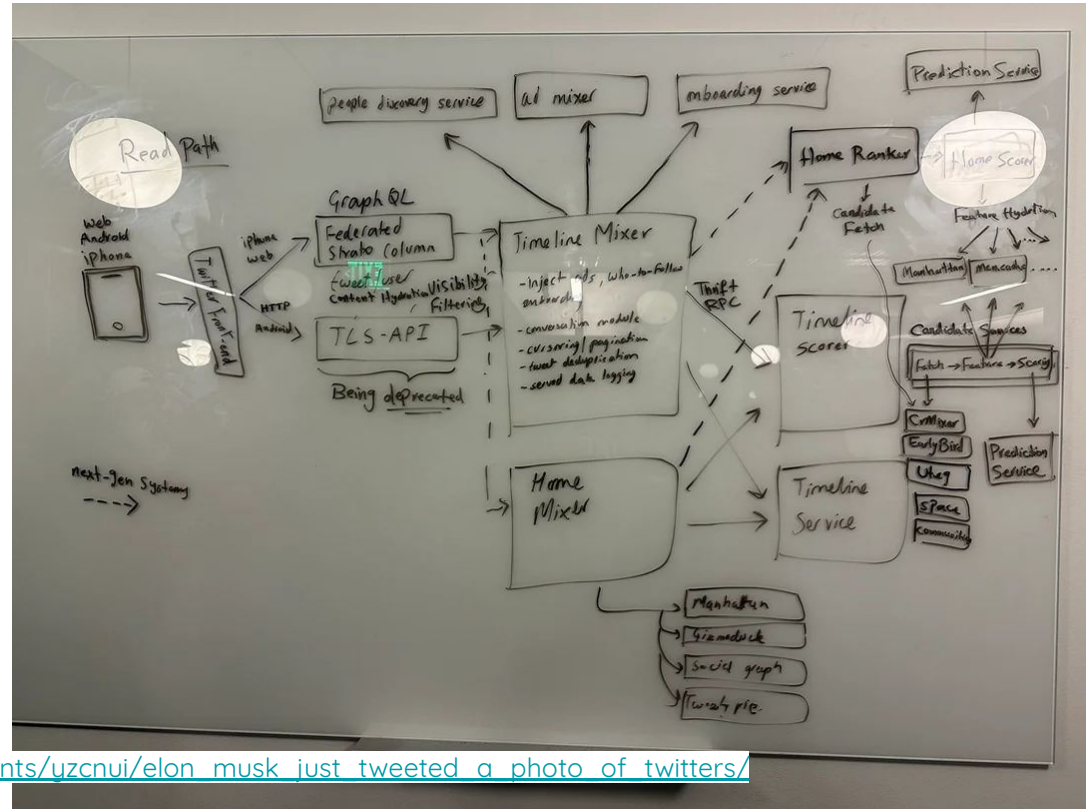
Class Activity: How would you design a new platform?

- Turn to person sitting beside you and discuss for the next 5-10 minutes:
- What type of content would it host?
 - Modality
 - Length
 - Format
 - ...
- What user base would you like to attract?
- How would you monetize it?
- What type of behavior would(n't) be allowed on the platform and how would you enforce it?

Today:

- 1. Data Collection 1: Web Scraping**
2. Application Programming Interfaces

How is the Platform Itself Created?



https://www.reddit.com/r/webdev/comments/yzcnu/elon_musk_just_tweeted_a_photo_of_twitters/

How is the Platform Itself Created? The Software

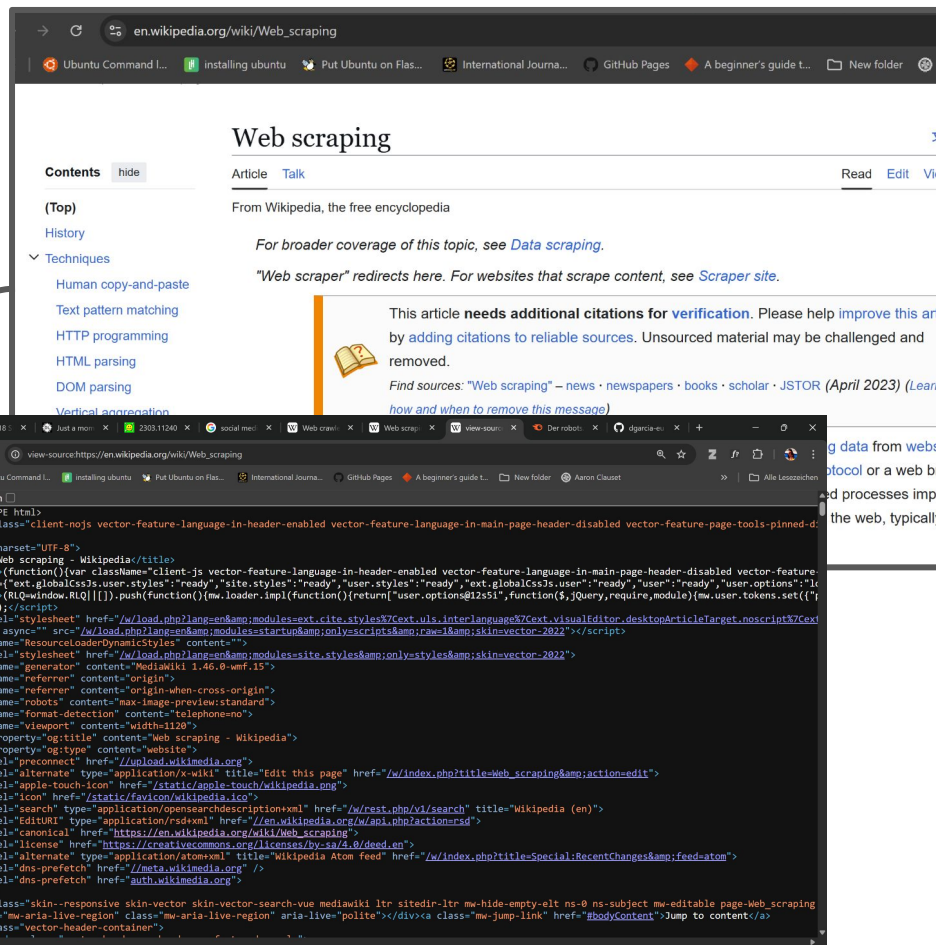
- A website and mobile apps where people interact - this is the "**front end**" that users see and tap on
- "**Back end**" handles all the actual work - storing posts, connecting friends, managing logins, etc.
- A **database** storing who all your users are, what they've posted, who's friends with whom, ...

How is the Platform Itself Created? The Hardware

- Data centers spread across different countries - you need these because millions of people are using your platform at once
- Servers that are always running, responding to requests like "show me my feed" or "upload this photo"

Web Scraping

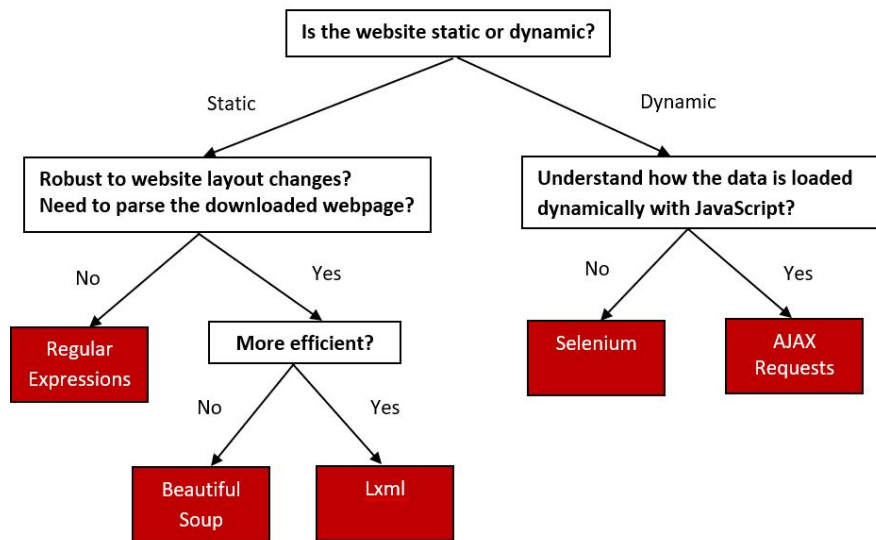
- Webpages are created using text-based markup languages, e.g., HTML (Hypertext Markup Language)
- **Web scraping:** collecting the code and content used to create a webpage
- You can do this manually (e.g., by inspecting the source), but ideally through code



view-source:https://en.wikipedia.org/wiki/Web_scraping

Web Scrapping

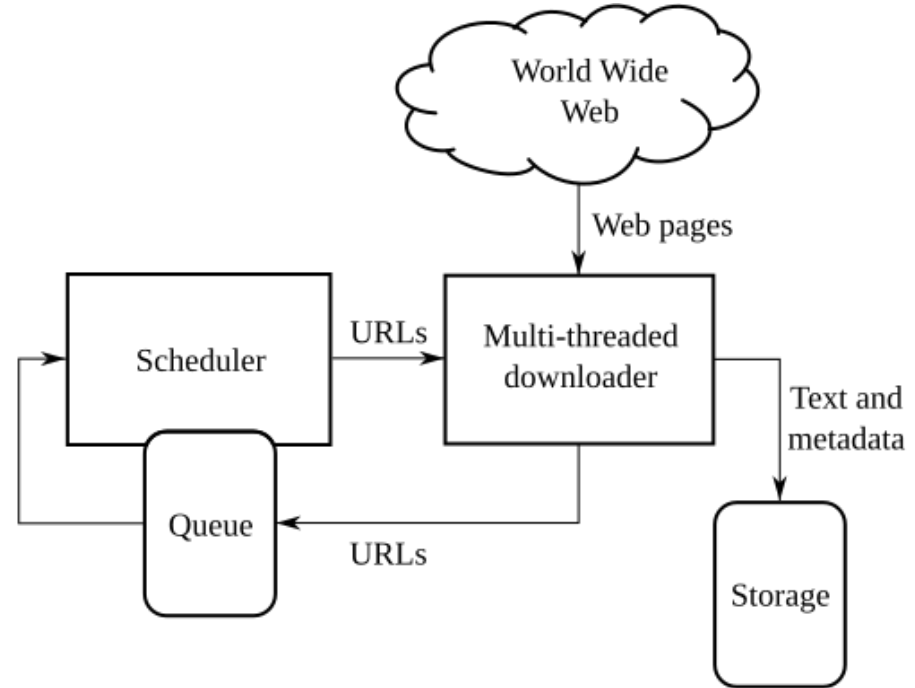
- For simple webpages, created with HTML and CSS, this is quite straightforward
- For dynamic webpages, more complicated
 - Most social media websites are dynamic — no two people see the same thing when they log in



<https://iqss.github.io/dss-webscrape/web-scraping-approaches.html>

Web Scraping vs. Web Crawling

- For collecting social media data, we don't just want to collect data from a single page
 - Multiple users' profiles from Twitter
 - Multiple subreddits on Reddit
- Thus you usually have a list of webpages that you need to scrape
- Therefore, you create crawlers that will do this process systematically



What's in the Platform vs. What you can Collect

- **What you see vs. what you get:** With web scraping, whatever we see on the platform should be accessible
 - This is not true for APIs (as we will see in the next lecture)
- **What you don't even see:** The platform itself is probably storing much more data than they are showing you
 - E.g., the active browsers with Wikipedia pages open
 - Lurkers and lurking-related activity
 - Private and direct messages
 - ...

Is Web Scraping Legal?

- Depends on the region, but general rule of thumb: web scraping is allowed for non-commercial purposes and research
 - In Germany: the German Federal Court of Justice (Bundesgerichtshof) ruled that the scraping of publicly available data, including personal data, might be permissible under German data protection laws.
 - However, the scraped data must not be used for any illegal purposes, and that the scraping must not violate the rights of the data subjects.
- Can you make web scraped data widely available?
 - Conflict between privacy and reproducible research

Ethical Considerations with Web Scraping

- Follow 'robots.txt' when possible
- Do not overload the website with requests
 - DoS: Denial of service attack
- Consider the pros and cons of research based on webscraping
 - Does it cause more benefit than harm?
 - Does it benefit or harm vulnerable people?



A screenshot of a web browser window displaying the content of the file `google.com/robots.txt`. The browser's address bar shows the URL `google.com/robots.txt`. The page content lists various directives for web crawlers, including `User-agent: *`, `Disallow: /search`, and several `Allow` and `Disallow` rules for specific paths like `/search/about`, `/search/static`, `/search/howsearchworks`, `/sdch`, `/groups`, `/index.html?`, `/?`, `/?hl=`, `/?hl=*g`, `/?hl=*g&gws_rd=ssl`, and `/?hl=*g&gws_rd=ssl`.

```
User-agent: *
Disallow: /search
Allow: /search/about
Allow: /search/static
Allow: /search/howsearchworks
Disallow: /sdch
Disallow: /groups
Disallow: /index.html?
Disallow: /?
Allow: /?hl=
Disallow: /?hl=*g
Allow: /?hl=*g&gws_rd=ssl
Disallow: /?hl=*g&gws_rd=ssl
```

<https://de.semrush.com/blog/robots-txt-leitfaden/>

Use Case: The Truth Social Dataset

- Twitter clone that became popular after Donald Trump was banned from several social media platforms
- Interesting platform to study a population that maynot be active on 'mainstream' platforms

Truth Social Dataset

Patrick Gerard, Nicholas Botzer, Tim Weninger

Department of Computer Science and Engineering
University of Notre Dame
Notre Dame, Indiana, USA
{pgerard2, nbotzer, tweninger}@nd.edu

Abstract

Formally announced to the public following former President Donald Trump's bans and suspensions from mainstream social networks in early 2022 after his role in the January 6 Capitol Riots, Truth Social was launched as an "alternative" social media platform that claims to be a refuge for free speech, offering a platform for those disaffected by the content moderation policies of the existing, mainstream social networks. The subsequent rise of Truth Social has been driven largely by hard-line supporters of the former president as well as those affected by the content moderation of other social networks. These distinct qualities combined with its status as the main mouthpiece of the former president positions Truth Social as a particularly influential social media platform and give rise to several research questions. However, outside of a handful of news reports, little is known about the new social media platform partially due to a lack of well-curated data. In the current work, we describe a dataset of over 823,000 posts to Truth Social and social network with over 454,000 distinct users. In addition to the dataset itself, we also present some basic analysis of its content, certain temporal features, and its network.

Introduction

The social media platform *Truth Social* was launched in February of 2022 about a year after the suspension of former United States President Donald Trump from Twitter, Facebook, and other social media platforms. Truth Social is largely stylized after Twitter where Tweets are instead called *Truths* and ReTweets are instead called *ReTruths*. Note that throughout the remainder of this paper we most commonly refer to *Truths* as *posts* in order to avoid confusion with the

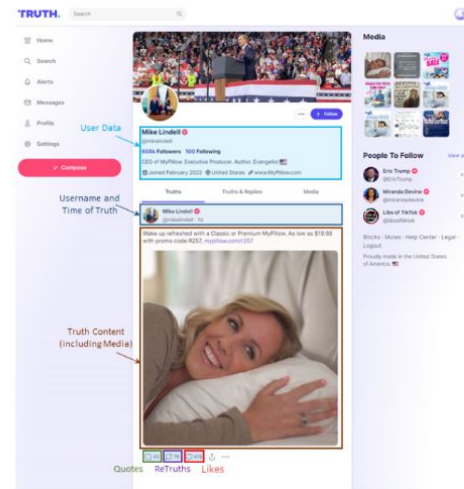


Figure 1: Annotated illustration of Truth Social Web Interface. The Web scraper extracted user data, and post information, including time, content (with links or other media), quotes, ReTruths, and likes.

Gerard, Patrick, Nicholas Botzer, and Tim Weninger. "[Truth social dataset](#)," Proceedings of the international AAAI conference on web and social media

Use Case: The Truth Social Dataset

- The authors use web crawling to collect data from Truth Social
- They use a ‘snowballing’ approach to collect data
- Do you think this approach affects the dataset quality in any way?

Dataset Collection Methodology

Collecting the posts and other activity data from Truth Social is not straightforward because the site does not provide a public API. Instead, we implemented a custom Web scraper and programmatically extracted the relevant context from Truth Social’s Web interface directly. This Web interface did not impose any crawling restrictions nor did it disallow any crawling with the `robots.txt` standard. Because no API was available the most straightforward way to collect posts was from each specific account.

We crawled the Truth Social one account at a time, starting with @realDonaldTrump and then iteratively in a breadth first manner over the followers of each account. Specifically, the crawling methodology proceeded as follows:

1. Collect information about the user’s follower count, following count, creation date.
2. Iterate through users following the user and users that the user follows, create an edge for each follower-followee relationship, and add that user to the breadth-first queue if that user has not been scraped in the past 14 days.
3. Extract the full set of available Truths posted by the user.

This crawl began on September 4, 2022 and continued until October 14, 2022. In that time, all content posted by 65,536 users was collected. The dataset, therefore, has the complete set of all posts for the visited users before September of 2022.

Announcement: Graded class discussions from next week (Lecture 4)

- Raise your hand and when I call on you, please say your name (at least for the first few sessions)
 - this applies to answering my questions or other's during the lecture
 - also for the class discussion at the end of each lecture
- Do not interrupt others
- How to prepare for the class discussions
 - Read/engage with the assigned readings
 - Summarize it in your own words in 2-3 sentences [don't use LLMs for the first summary, but feel free to brainstorm with one **after** you've read it yourself]
 - For papers, think about its strengths and limitations
 - Reminder: [How to read a research paper](#)
- Project Information will be available in Lecture 5, but feel free to ask us questions about it in the meantime

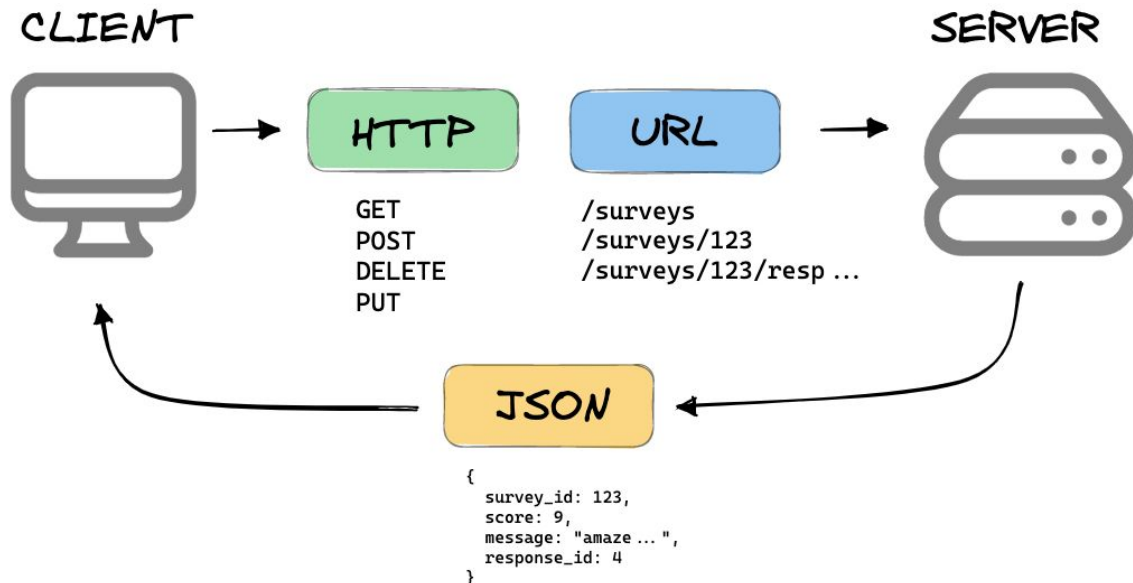
Today:

1. Data Collection 1: Web Scraping
2. **Application Programming Interfaces**

APIs

- Application Programming Interface
- Typically not designed for research purposes (therefore not tailored to research needs)
- Several platforms offer APIs as access points
- Often you have to register, sometimes also accept terms of services

WHAT IS A REST API?



APIs are typically not designed for research

- Core Purpose of Social Media APIs
 - Allow developers to interact with platform features
 - Expand the platform's ecosystem beyond its own website or app, e.g., [Reddit Mod tools](#),
 - Social login (“Sign in with Facebook”) and sharing content directly from other apps
 - Social media management tools (e.g., schedulers)
 - ...

APIs are typically not designed for research

- Core Purpose of Social Media APIs
 - Allow developers to interact with platform features
 - Expand the platform's ecosystem beyond its own website or app, e.g., [Reddit Mod tools](#),
 - Social login (“Sign in with Facebook”) and sharing content directly from other apps
 - Social media management tools (e.g., schedulers)
 - ...
- To use an **official** social media API, you must:
 - Create an account
 - Register as a developer
 - Create an app
 - Authenticate users
 - Obtain access tokens — **And be careful not to share them with anyone!**
 - Respect permissions & rate limits

Create app for extracting Reddit data

1. Go to the link in the Notebook (<https://old.reddit.com/prefs/apps/>)
2. Create Reddit Account

3. Go to Apps

4. Create App (select script, on the URL you can put anything you like)

4. Create App (select script, on the URL you can put anything you like)

create application

By creating an app, you agree to Reddit's Developer Terms and Data Api Terms. You must also register to use the API.

5. Make A NOTE of personal use script and secret tokens (you will need them for creating a reddit instance)

Example: Twitter

- Different access options for different purposes:
 - Twitter Academic API:
<https://developer.twitter.com/en/products/twitter-api/academic-research>
 - Twitter Developer: <https://developer.twitter.com/>
 - premium "Firehose" access. See also Twitter Enterprise:
<https://developer.twitter.com/en/enterprise>
 - Changes to APIs policies over time (functionalities, user agreements)
 - Limitations on volume and functions
- IMPORTANT: – Free APIs cover 7 days tweets; Premium APIs exist for 30-day search and full archive search.

Example: Twitter

- Different access options
 - Twitter Academic / <https://developer.twitter.com/en/research>
 - Twitter Developer: premium "Firehose" <https://developer.twitter.com/en/enterprise/twtrfirehose>
- Changes in APIs policies (functionalities, user agreements, etc.)
- Limitations on volume (IMPORTANT: – Free API 30-day search and full

RIP Twitter API: A Eulogy to Its Vast Research Contributions

Ryan Murtfeldt, Sejin Paik, Naomi Alterman, Ihsan Kahveci, Jevin D. West

Abstract

Since 2006, Twitter's APIs have been rich sources of data for researchers studying social phenomena such as misinformation, public communication, crisis response, and political behavior. However, in 2023, Twitter began heavily restricting data access, dismantling its academic access program, and setting the Enterprise API price at \$42,000 per month. Lacking funds to pay this fee, academics are scrambling to continue their research. This study systematically tabulates the number of studies, citations, publication dates, disciplines, and major topics of research using Twitter data between 2006 and 2024. While we cannot know exactly what will be lost now that Twitter data is cost-prohibitive, we can illustrate its research value during the years it was available. A search of eight databases found that between 2006 and 2024, a total of 33,306 studies were published in 8,914 venues, with 610,738 citations across 16 disciplines. Major disciplines include social science, engineering, data science, and public health. Major topics include information dissemination, tweet credibility, research methodologies, event detection, and human behavior. Twitter-based studies increased by a median of 25% annually from 2006 to 2023, but following Twitter's decision to charge for data, the number of studies dropped by 13%. Much of the 2024 research likely used data collected

More examples: APIs

- Facebook for Developers: <https://developers.facebook.com/>
- Facebook Ads API:
<https://developers.facebook.com/docs/marketing-apis/>
- Instagram Developer: <https://www.instagram.com/developer/>
- YouTube Developers: <https://developers.google.com/youtube/>
- Weibo API:
<http://open.weibo.com/wiki/API%E6%96%87%E6%A1%A3/en>

More examples: APIs

- Tiktok: <https://github.com/dfreelon/pyktok>
- 4chan: <https://github.com/4chan/4chan-API>
- Gab: <https://github.com/a-tal/gab>

APIs vs Web Scraping

Category	Web Scraping	API-Based Data Collection
Access Method	Extracts data from website front-end (HTML pages)	Sends structured requests to official backend endpoints
Official Support	Not officially supported by platforms	Officially provided by platforms like Facebook, Instagram, X, and LinkedIn* Unofficial APIs also exist but much less stable
Data Format	Unstructured HTML → requires parsing	Structured JSON/XML → ready to use
Stability	Breaks when website layout changes	Stable and version-controlled
Authentication	May bypass login systems	Requires secure login (OAuth, tokens)
Permissions	No formal permission model	Uses scoped access permissions
Rate Limits	No official limits, but risk of blocking	Clearly defined request limits
Legal Risk	Potential Terms of Service violations	Governed by developer agreements
Maintenance Effort	High – frequent updates required	Lower – documented and supported
Ethical Considerations	May raise privacy concerns	Designed with privacy safeguards

Research that APIs have enabled

OPEN

Evaluating the effect of viral posts on social media engagement

Emanuele Sangiorgio^{1,2}, Nicolò Di Marco², Gabriele Etta², Matteo Cinelli², Roy Cerqueti^{1,2} & Walter Quattrocchi²

As virality has become increasingly central in shaping information sources' strategies, it raises concerns about its consequences for society, particularly when referring to the impact of viral news on the public discourse. Nonetheless, there has been little consideration of whether these viral events genuinely boost the attention received by the source. To address this gap, we analyze content timelines from over 1000 European news outlets from 2018 to 2023 on Facebook and YouTube, employing a Bayesian structural time series model to evaluate the impact of viral posts. Our results show that most viral events do not significantly increase engagement and rarely lead to sustained growth. We identify two primary types of viral effects corresponding to different mechanisms of collective attention response. A 'loaded-type' virality manifests after a sustained growth phase, representing its final burst, followed by a decline in attention. A 'sudden-type' virality, with news emerging unexpectedly, reactivates the collective response process. Moreover, quick viral effects fade faster, while slower processes lead to more persistent growth. These findings highlight the transient nature of viral events and underscore the importance of consistent, steady attention-building strategies to establish a solid connection with the user base rather than relying on sudden visibility spikes.

Keywords Social media, Virality, Attention economy

The advent and proliferation of social media have fundamentally altered the information landscape^{1,2}. As these platforms have become integrated into our daily lives, transforming into essential tools for information diffusion^{3,4} and personal communication⁵, they have merged entertainment-driven business models with complex social dynamics⁶, raising significant concerns about their potential impact on social dynamics^{7–11}. Offering unprecedented opportunities for content to achieve rapid and widespread attention^{12–17}, social media have become crucial environments for the spread of information and misinformation worldwide^{18–21}, especially during sensitive periods such as global elections^{22–27}.

This complex interplay has resulted in an ecosystem in which information overload is the foremost feature¹⁸, considerably complicating the search for information while giving rise to the phenomenon of infodemics^{18,20}, such as for the COVID-19 pandemic. To this effect, evidence in the literature highlights how online users are prone to consume information aligning with their existing beliefs^{21–25} and commonly ignore opposing viewpoints^{26,27}. In this competitive environment, which stands out for such an unprecedented amount of available content, a wide range of content creators—from news organizations to individual influencers^{28,29}—compete for the limited resource that is users' attention^{30–33}.

Understanding the attention economy in the digital domain is paramount for navigating this vying information market, whereby the pursuit of virality³⁴, characterized by content's exceptional reach and engagement, plays a pivotal role in shaping how information sources design their strategies of production and diffusion of content. Though one of the most complex issues is understanding why certain items achieve sudden and widespread

Sangiorgio, Emanuele, et al. "[Evaluating the effect of viral posts on social media engagement](#)." *Scientific Reports* 15.1 (2025): 639. [DOI: 10.1038/s41598-025-00000-0](#)

Measuring exposure to misinformation from political elites on Twitter

Received: 4 January 2022

Mohsen Mosleh^{1,2,3} & David G. Rand^{1,4,5}

Accepted: 7 November 2022

Published online: 21 November 2022

Check for updates

Misinformation can come directly from public figures and organizations (referred to here as "elites"). Here, we develop a tool for measuring Twitter users' exposure to misinformation from elites based on the public figures and organizations they choose to follow. Using a database of professional fact-checks by PolitiFact, we calculate falsity scores for 816 elites based on the veracity of their statements. We then assign users an elite misinformation-exposure score based on the falsity scores of the elites they follow on Twitter. Users' misinformation-exposure scores are negatively correlated with the quality of news they share themselves, and positively correlated with estimated conservative ideology. Additionally, we analyze the co-follower, co-share, and co-retweet networks of 5000 Twitter users and observe an association between conservative ideology and misinformation exposure. Finally, we find that estimated ideological extremity is associated with more misinformation exposure to a greater extent for users estimated to be conservative than for users estimated to be liberal. Finally, we create an open-source R library and an Application Programming Interface (API) making our elite misinformation-exposure estimation tool openly available to the community.

There has been growing public concern about misinformation on social media. Accordingly, a great deal of effort has been invested by practitioners and researchers into investigating the spread of online misinformation¹. Prior work has largely focused on belief in, and sharing of, articles (often headlines) from reliable versus unreliable news domains. While this kind of misinformation was a particular focus of interest in the wake of the 2016 election cycle (e.g., far-fetched headlines from little-known publishers going viral), a different form of misinformation has begun to gain attention in recent years: coordinated misinformation campaigns orchestrated by public figures and organizations (i.e., referred to here as "elites"), such as the claims of widespread fraud in the 2020 U.S. Presidential Election made by Republican politicians². Given the large body of evidence documenting the impact of political elites on voter attitudes and behaviors^{3–7}, exploring exposure to elite-based misinformation (rather than just news domains) is important.

Furthermore, in focusing on what people believe and share, prior work has largely overlooked what (mis)information people are exposed to (a notable exception is ref. 8). Although exposure and sharing are obviously related (inasmuch as you can only share content that you are exposed to), they are fundamentally different constructs. Most people share only a tiny fraction of the content they are exposed to⁹, and therefore examining the content someone shares provides a very limited picture of a person's information environment. The choice of whom to follow (and thus what information to expose oneself to) is particularly important in light of evidence that simply being exposed to content, even if it is highly implausible, makes it subsequently seem truer¹⁰.

Here, we introduce an approach for studying misinformation on social media that specifically focuses on exposure to misinformation from elites (defined as public figures and organizations). In particular, we estimate Twitter users' exposure to misinformation from elites by

¹Management Department, University of Exeter Business School, Exeter, UK. ²Alan Turing Institute, London, UK. ³Sloan School of Management, Massachusetts Institute of Technology, Cambridge, MA, USA. ⁴Initiative on the Digital Economy, Massachusetts Institute of Technology, Cambridge, MA, USA. ⁵Department of Brain and Cognitive Sciences, Massachusetts Institute of Technology, Cambridge, MA, USA. ✉e-mail: m.mosleh@exeter.ac.uk

Mosleh, Mohsen, and David G. Rand. "[Measuring exposure to misinformation from political elites on Twitter](#)." *Nature Communications* 13.1 (2022): 7144.

Let's try some examples

- Unofficial reddit API: <https://arctic-shift.photon-reddit.com/search>
- https://github.com/ArthurHeitmann/arctic_shift

APIs aren't perfect for research

- Previously Twitter used to provide a regular streaming API and a **firehose** API
- Streaming API: at most a 1% sample of all the data → free
- Firehose API: 100% data → paid, now costs 42KUSD
- No guarantee that the 1% is a representative sample of all tweets

Morstatter, Fred, et al. "[Is the sample good enough? Comparing data from Twitter's streaming API with Twitter's firehose.](#)" *Proceedings of the international AAAI conference on web and social media*. Vol. 7. No. 1. 2013.

Is the Sample Good Enough? Comparing Data from Twitter's Streaming API with Twitter's Firehose

Fred Morstatter
Arizona State University
699 S. Mill Ave.
Tempe, AZ, 85281

Jürgen Pfeffer
Carnegie Mellon University
5000 Forbes Ave.
Pittsburgh, PA, 15213

Huan Liu
Arizona State University
699 S. Mill Ave.
Tempe, AZ, 85281

Kathleen M. Carley
Carnegie Mellon University
5000 Forbes Ave.
Pittsburgh, PA, 15213

Abstract

Twitter is a social media giant famous for the exchange of short, 140-character messages called "tweets". In the scientific community, the microblogging site is known for openness in sharing its data. It provides a glance into its millions of users and billions of tweets through a "Streaming API" which provides a sample of all tweets matching some parameters preset by the API user. The API service has been used by many researchers, companies, and governmental institutions that want to extract knowledge in accordance with a diverse array of questions pertaining to social media. The essential drawback of the Twitter API is the lack of documentation concerning what and how much data users get. This leads researchers to question whether the sampled data is a valid representation of the overall activity on Twitter. In this work we embark on answering this question by comparing data collected using Twitter's sampled API service

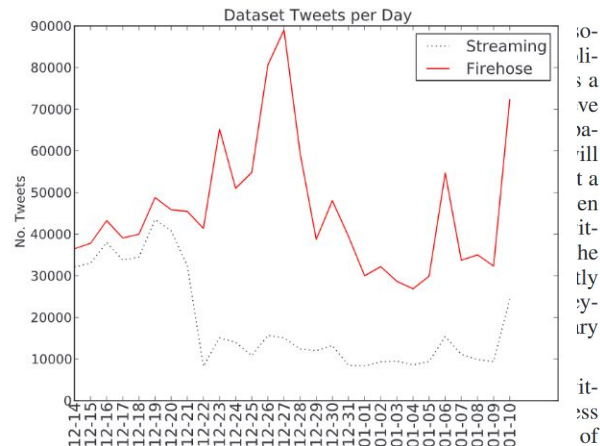


Figure 2: Raw tweet counts for each day from both the Streaming API and the Firehose.

APIs are forever changing!

Computational research in the post-API age

Deen Freelon

University of North Carolina at Chapel Hill

Forthcoming in *Political Communication*

Keywords: API, computational, Facebook, Twitter, social media

2018-08-20

On April 4, 2018, the post-API age reached a milestone. On that day, Facebook closed access to its Pages API, which had allowed researchers to extract all posts, comments, and associated metadata from public Facebook pages (Schroepfer, 2018). This decision followed the company's April 2015 closure of its public search API, which provided searchable access to all public posts within a rolling two-week window (Facebook, n.d.). The closure of the Pages API eliminated all terms of service (TOS)-compliant

Freelon, Deen. "[Computational research in the post-API age.](#)" *Political Communication* 35.4 (2018): 665-668.

SM+S
social media + society

Below the Radar: Private Groups, Locked Platforms and Ephemeral Content

Where Have All the Data Gone? A Critical Reflection on Academic Digital Research in the Post-API Age

Rebekah Tromble

Abstract

In the wake of the 2018 Facebook–Cambridge Analytica scandal, social media companies began restricting academic researchers' access to the easiest, most reliable means of systematic data collection via their application programming interfaces (APIs). Although these restrictions have been decried widely by digital researchers, in this essay, I argue that relatively little has changed. The underlying relationship between researchers, the platforms, and digital data remains largely the same. The platforms and their APIs have always been proprietary black boxes, never intended for scholarly use. Even when researchers could mine data seemingly endlessly, we rarely knew what type or quality of data were at hand. Moreover, the largesse of the API era allowed many researchers to conduct their work with little regard for the rigor, ethics, or focus on societal value, we should expect from scholarly inquiry. In other words, our digital research processes and output have not always occupied the high ground. Rather than viewing 2018 and Cambridge Analytica as a profound disjuncture and loss, I suggest that digital researchers need to take a more critical look at how our community collected and analyzed data when it still seemed so plentiful, and use these reflections to inform our approaches going forward.

Keywords

digital data, APIs, social media research, research ethics

Introduction

When the Facebook–Cambridge Analytica (CA) scandal burst into the headlines in March 2018, it cast a spotlight on tech companies' uses and abuses of personal data (Cadwalladr & Graham-Harrison, 2018). For years, platforms such as Facebook and Google had been gobbling up user data, tracking the minutiae of our daily lives, and selling or sharing that information with companies that desired our attention. Whether these companies sought to influence our consumer

included academics. From anthropology to psychology, economics to health science, scholars from a wide variety of disciplines relied on these APIs to gather large amounts of data for research into the content and behaviors found in digital spaces, and the CA-inspired restrictions significantly undermined multiple lines of research (Freelon, 2018; Hemsley, 2019).

Not surprisingly, digital researchers have reacted to this new "post-API age" (Freelon, 2018), with a mixture of frustration, as the platforms took away the tools that had made their research possible. The impact of this has been to restrict, or shut down, the flow of data to scientific knowledge. Digital research provides

Tromble, Rebekah. "[Where have all the data gone? A critical reflection on academic digital research in the post-API age.](#)" *Social Media+ Society* 7.1 (2021): 2056305121988929.

APIs are forever changing!

- For example, Facebook completely closed down many of its APIs and it is now very hard to get Facebook data besides FB Ads
- Twitter's API has undergone substantial changes over time
- Conclusion: Have to stay vigilant and continuously update our code to keep up with the APIs

Ethical Considerations with API-based Data Collection

- Similar to web scraping, respect terms of service (ToS)
- Do not create multiple accounts and keys under different names
- Bigger questions about storing and openly releasing data from APIs
 - Should you retroactively delete everything that was deleted on the platform?
 - Sometime adhering to ToS => less reproducibility

The End of the Rehydration Era The Problem of Sharing Harmful Twitter Research Data

Dennis Assenmacher, Indira Sen, Leon Fröhling, Claudia Wagner

GESIS - Leibniz Institute for the Social Sciences

{dennis.assenmacher, indira.sen, leon.froehling, claudia.wagner}@gesis.org

Abstract

Social media research is currently confronted with a data-sharing problem, as social media platforms prohibit full data distribution in their terms of service. Until recent changes to the platform, Twitter was an exception, allowing academics to legally share Tweet and user IDs with peers, which could then be re-collected using the Academic API endpoints. This work investigates how Twitter data is currently shared in two domains of harmful online communication — abusive language and social bot detection. We find that the currently frequently utilized intermediate strategy of sharing Twitter IDs suffers from substantial data loss, leading to the incomparability of computational results. Moreover, recent changes in the API result in additional expenses and an increased collection time that may have an impact on the feasibility of research projects. All of these aspects further fuel the reproducibility crisis that social media analytics currently faces. To improve the current situation, we propose several best practices for research projects utilizing ID-based datasets for their experiments and provide recommendations for researchers who want to share their Twitter data with peers.

Introduction

Computational social scientists are concerned with investigating social and socio-technical phenomena through the utilization of computational methods and digital trace data, often from large social media platforms. Despite millions of new data instances being produced on these platforms within seconds, data access and distribution are currently inherently restricted by the platform's Terms of Services (ToS)¹. Since some platforms are more restrictive than others, we currently observe an inherent platform bias in social media analytics (Tufekci 2014). Most of the current research endeavors in different Computational Social Science sub-domains, such

the real-time stream of content posted on Twitter. However, according to Twitter's ToS, it is not allowed to share full data collections of these observations with other researchers publicly.

As discussed in Assenmacher et al. (2021), this leads to an inherent problem of limited reproducibility for research that relies on this type of social media data. Different approaches are prevalent in the community to circumvent the data-sharing problem (e.g., access via request). One method stands out in this context, especially for Twitter: *Rehydration*. Instead of sharing Tweets and associated meta-data directly, only unique Tweet identifiers (IDs) are shared. Using lists of IDs, the original Tweets and user objects can be retrieved from the Twitter API, allowing researchers to recreate the original data collection in theory. However, Twitter provides no access to Tweets or user profiles that have been removed from the platform (by the users themselves or removed by Twitter, e.g., because of a ToS violation). Investigating harmful online communication patterns in social media leads to a situation where a substantial amount of observations will become inaccessible over time, as this content is prone to be removed from the platform, either by the platform itself (moderation) or user-initiated deletion. When it comes to the evaluation of new methodological approaches (e.g., detection mechanisms for abusive language) and the comparison of empirical findings on social or socio-technical phenomena (e.g., fake news), this does not make for an acceptable status quo as it inherently hinders the reproducibility of research that is based on social media data. Recently - after Elon Musk's acquisition of Twitter - the platform announced fundamental changes to its APIs, including the Academic one.² ID-based Tweet retrieval will be strictly limited to 10,000 Tweets per month at a price-point of \$100, introducing even more practical prob-

Assenmacher, Dennis, et al. "[The end of the rehydration era-the problem of sharing harmful twitter research data.](#)" *2nd Workshop on Novel Evaluation Approaches for Text Classification Systems (NEATCLaS)*. 2023.

Summary and Next Week

- This session we learned:
 - Data Collection 2: Application Programming Interfaces (API)
 - Official and Unofficial APIs
 - Why Platforms offer APIs
- Next time:
 - Data Collection 3: Platform Affordances — how do the design of platforms affect behavior and social media data analysis?
 - Readings:
 - Hase, Valerie, Karin Boczek, and Michael Scharkow. "[Adapting to affordances and audiences? A cross-platform, multi-modal analysis of the platformization of news on Facebook, Instagram, TikTok, and Twitter.](#)" Digital Journalism 11.8 (2023): 1499-1520.
 - Guinaudeau, Benjamin, Kevin Munger, and Fabio Votta. "[Fifteen seconds of fame: TikTok and the supply side of social video.](#)" Computational communication research 4.2 (2022): 463-485.

Additional Resources

- Nicola Osborne, Univ. of Edinburgh: Working with Social Media Data: Ethics & good practice around collecting, using and storing data
<https://www.slideshare.net/suchprettyeyes/working-with-social-media-data-ethics-good-practice-around-collecting-using-and-storing-data>
- Roberto Ulloa, GESIS: Introduction to Online Data Acquisition
<https://www.youtube.com/watch?v=inUvEFLG5EA>
- Deubel, A., Breuer, J., Weller, K. (2023). Collecting Social Media Data: Tools for Obtaining Data from Social Media Platforms. Navigating Research Data and Methods, Vol. 1. CAIS Center for Advanced Internet Studies.
<https://www.cais-research.de/wp-content/uploads/Collecting-Social-Media-Data.pdf>

Questions?